Example 5.19 Find the sum: $1 + \frac{4}{5} + \frac{7}{25} + \frac{10}{125} + \cdots$.

Solution:
Here $a = 1$, $d = 3$ and $r = \frac{1}{5}$.

$$\begin{aligned} S_\infty &= \frac{a}{1-r} + \frac{dr}{(1-r)^2} \\ &= \frac{1}{1-\frac{1}{5}} + \frac{3 \times \frac{1}{5}}{(1-\frac{1}{5})^2} \\ &= \frac{5}{4} + \left(\frac{3}{5}\right)\left(\frac{25}{16}\right) \\ &= \frac{35}{16} \end{aligned}$$

5.6.4 Telescopic Summation for Infinite Series

We discussed about summing terms of a finite sequence using telescopic summation technique in Section 5.5.2. The same applies for infinite series also.

Example 5.20 Find $\sum\limits_{n=1}^{\infty} \frac{1}{n^2+5n+6}$.

Solution:
Let a_n denote the nth term of the given series. Then $a_n = \frac{1}{n^2+5n+6}$. By using partial fraction, we get

$$a_n = \frac{1}{n+2} - \frac{1}{n+3}.$$

Let s_n denote the sum of first n terms of the given series. Then

$$s_n = a_1 + a_2 + \cdots + a_n = \left(\frac{1}{3} - \frac{1}{4}\right) + \left(\frac{1}{4} - \frac{1}{5}\right) + \left(\frac{1}{5} - \frac{1}{6}\right) + \cdots + \left(\frac{1}{n+2} - \frac{1}{n+3}\right) = \frac{1}{3} - \frac{1}{n+3}.$$

But as n tends to infinity, $\frac{1}{n+3}$ tends to zero and hence $\frac{1}{3} - \frac{1}{n+3}$ tends to $\frac{1}{3}$. In other words s_n tends to $\frac{1}{3}$. Thus $\sum\limits_{n=1}^{\infty} \frac{1}{n^2+5n+6} = \frac{1}{3}$.

5.6.5 Binomial Series

In the discussion on geometric series we have seen that

$$\frac{1}{1-x} = 1 + x + x^2 + \cdots, \frac{1}{1+x} = 1 - x + x^2 - \cdots \text{ and } \frac{1}{1-2x} = 1 + 2x + 4x^2 + \cdots$$

for some suitable values of x. But the expressions $\frac{1}{1-x}$, $\frac{1}{1+x}$ and $\frac{1}{1-2x}$ can be written as $(1-x)^{-1}$, $(1+x)^{-1}$ and $(1-2x)^{-1}$. This suggests us a possibility of having negative exponents, that is negative powers, for $(1+x)$, $(1-x)$ and so on. Yes. This is possible. We can have any power, positive or negative, integer or rational. We can even have irrational exponent for $(1+x)$. We already proved